

Git is a version control system it tracks change in files and helps us to know what changes we have been doing

so now to make git to track a folder we have to initialize a repository

Command : git init

This creates a hidden .git directory

Command : ls -a

So the folder which has been created by the above command stores, commits, history, branches and configurations so this folder is a git repository so if this folder gets deleted then the entire project history is gone

To Clear everything on screen

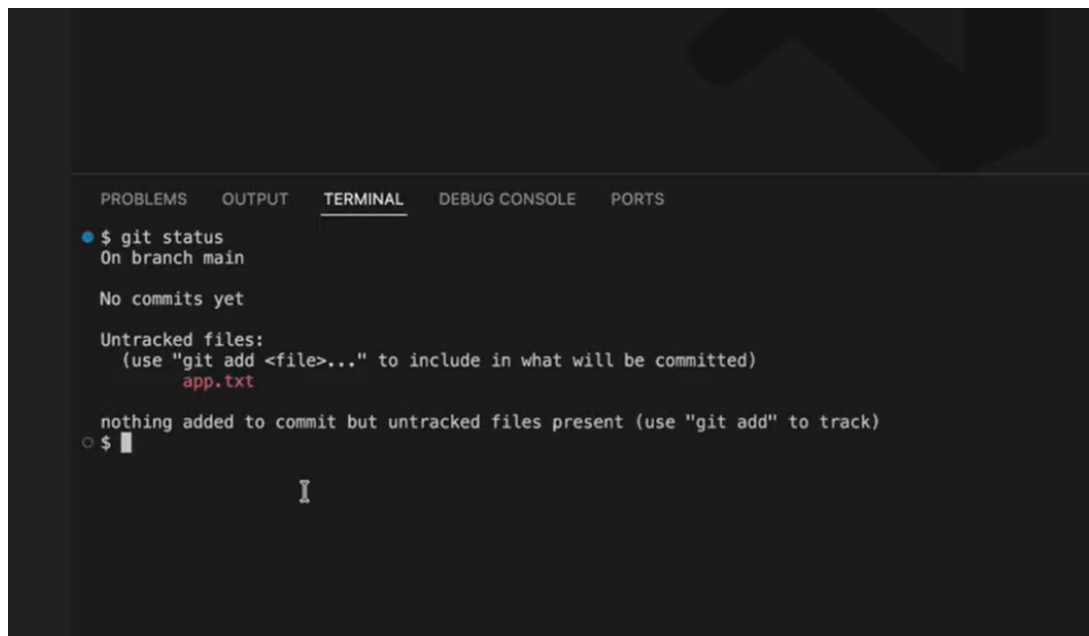
Command : clear

To create a simple file

Command : echo "String to give comment to your file" > app.txt >> This is the name of your file (app)

To check the status

Command : git status



```
PROBLEMS  OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  app.txt

nothing added to commit but untracked files present (use "git add" to track)
$
```

So this error means git can see the file app.txt which is created but it cannot commit to it

So before knowing commit

We need to know that git works in 3 areas

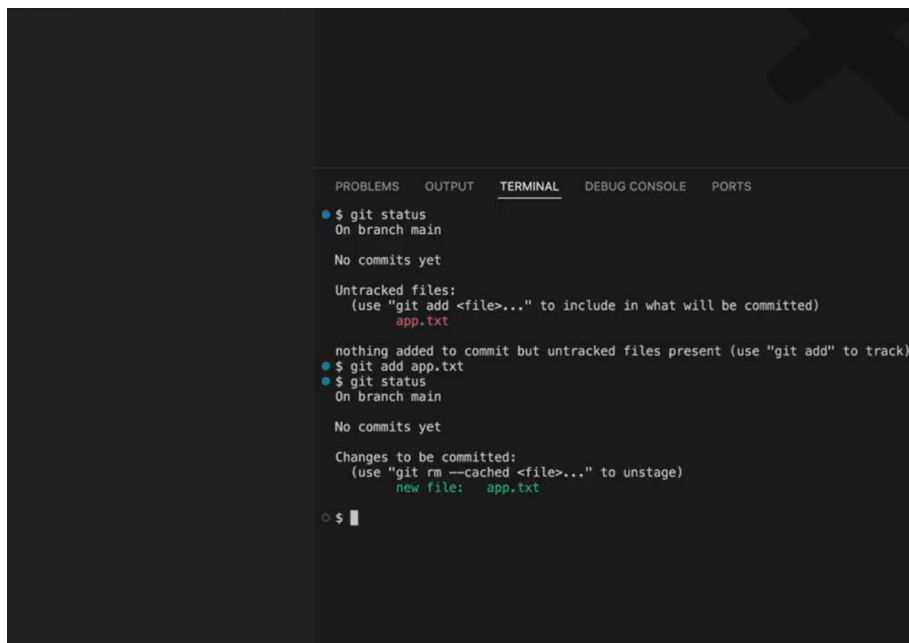
1. Working directory
2. Staging area
3. Commit History

Working Directory means this is where we create and edit our files

So app.txt which has been created is in working directory

Staging area this is where preparation is done

Command: git add app.txt so to check the status use the status command

A terminal window with tabs for PROBLEMS, OUTPUT, TERMINAL, DEBUG CONSOLE, and PORTS. The TERMINAL tab is active. The output shows the following commands and their results:

```
$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    app.txt

nothing added to commit but untracked files present (use "git add" to track)
$ git add app.txt
$ git status
On branch main

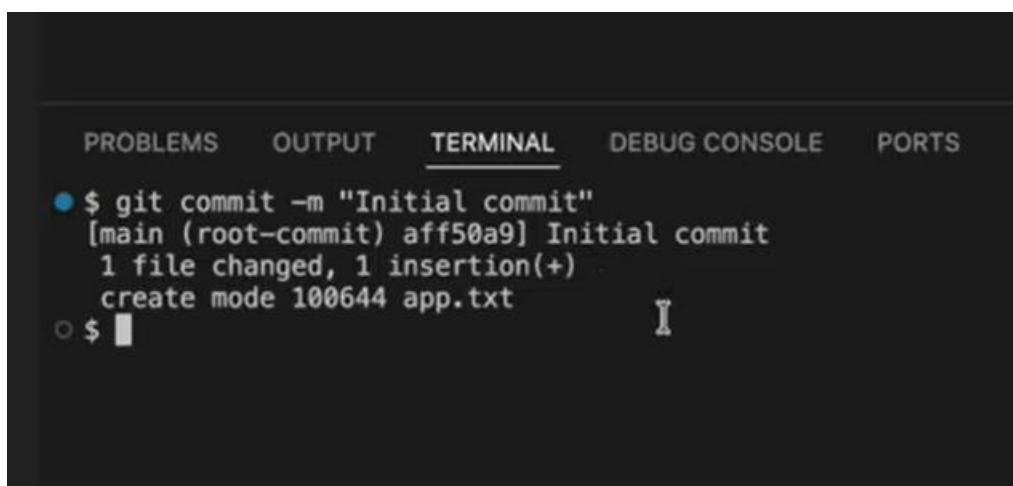
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   app.txt
```

Git add command just stages the file to commit but it does not create a commit
it only prepares the changes for the next commit

make git to commit

Command: git commit -m "String"

A terminal window with tabs for PROBLEMS, OUTPUT, TERMINAL, DEBUG CONSOLE, and PORTS. The TERMINAL tab is active. The output shows the following command and its result:

```
$ git commit -m "Initial commit"
[main (root-commit) aff50a9] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 app.txt
```

It has been committed now

A Commit is a snapshot of our project at this moment it's the first snapshot that is why it says initial commit

To view the history

Command: git log --online



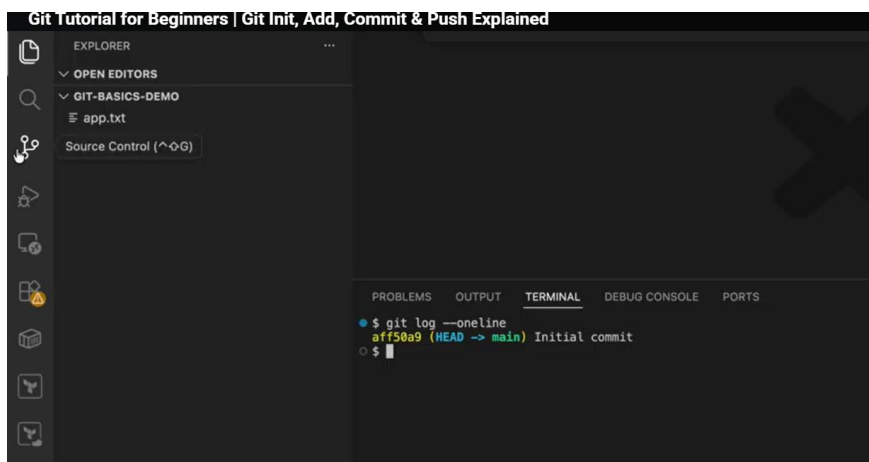
```
PROBLEMS  OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS

$ git log --online
aff50a9 (HEAD -> main) Initial commit
$
```

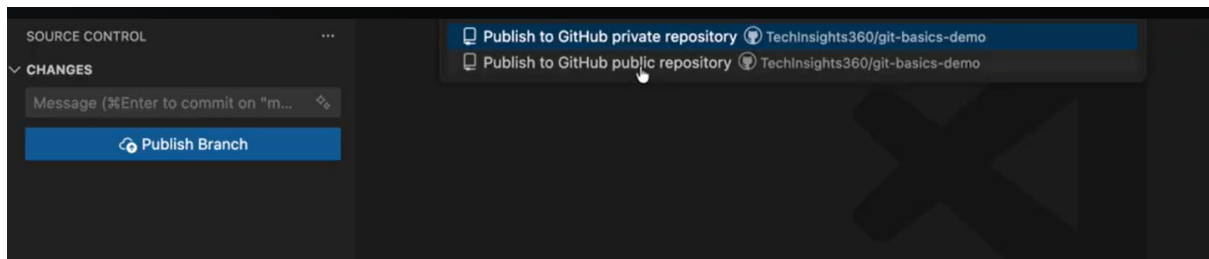
So till now whatever we made is in local machine not in the github

To push it to github we can use vscode and for doing it firstly in git

Goto Source control



Click on publish branch and choose public or private



Then goto your github profile in the internet and goto repositories you will have the file which we have created

Next steps she created a new folder then staged and pushed the file

Push Command: git push

```
PROBLEMS  OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS

• $ echo "Second file for Git demo" > notes.txt
• $ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

...
nothing added to commit but untracked files present (use "git add" to track)
• $ git add notes.txt
• $ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   notes.txt

• $ git commit -m "Add notes file"
[main d0f6773] Add notes file
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
• $ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 14 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 313 bytes | 313.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/TechInsights360/git-basics-demo.git
   aff50a9..d0f6773  main -> main
○ $
```